Exercise:2.2. Create a 3D Solid Model from an Isometric Drawing

Objective:

To model a 3D solid part from an isometric engineering drawing using **Autodesk Fusion 360**.

System Requirements:

1. **Software:** Autodesk Fusion 360
2. **OS:** Windows 11
3. **Memory:** 16 GB RAM recommended (Integrated graphics with 1 GB+ recommended)
4. **Internet:** 2.5 Mbps download / 500 Kbps upload or faster

Step-by-Step Modeling Procedure:

Step 1: Base Sketch & Extrusion (XY Plane)

1. **Select Plane:** Choose the **Top (XY) Plane** to start your initial sketch.
2. **Base Rectangle:**
3. Navigate to **Sketch > Rectangle > Center Rectangle**, snapping to the origin.
   Set dimensions: **56 mm width** and **18 mm height** (press **Tab** between fields).
4. **End Arcs & Circles:**
5. Add **Center Point Arcs** at both ends with a radius of **R 18 mm**.
   Place **Center Point Circles** on both sides with a diameter of **dia 16 mm**.
6. **Finish Sketch:** Click **Finish Sketch**.
7. **Extrude Base:**
   Select **Extrude**, pick the sketch profile, enter a distance of **8 mm**, and press **Enter**.

Step 2: Second Tier Feature

1. **New Sketch:** Select **Create Sketch** and click on the **top surface** of the existing 3D block.
2. **Rectangle Profile:**
3. Go to **Sketch > Rectangle > Center Rectangle** from the origin.
   Set dimensions to **40 mm** (width) and **10 mm** (height). Finish the sketch.
4. **Extrude Feature:**
5. Select **Extrude**, choose the new rectangle profile, set the distance to **32 mm**, and press **Enter**.

Step 3: Triangular Feature

1. **Triangle Profile:**
2. Create a new sketch on the working plane.
   Go to **Sketch > Triangle > Center Triangle** centered at the origin.
   Set dimensions to **32 mm** (width) and **12 mm** (height). Finish the sketch.

3. **Symmetric Extrude:**
   Select **Extrude**, pick the triangle sketch, set the distance to **6 mm**, and choose the **Symmetric / Both Directions** option. Press **Enter**.

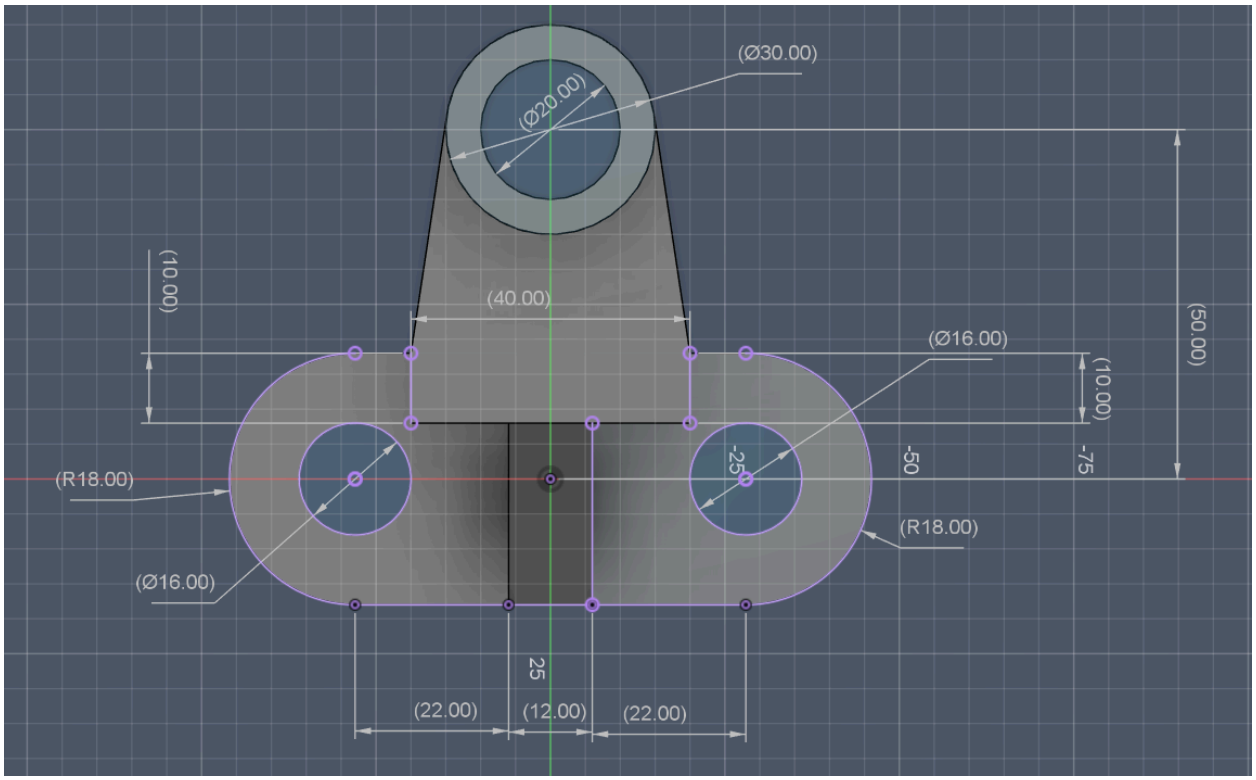
Step 4: Polygon Feature

1. **Polygon Profile:**
2. Go to **Sketch > Polygon > Circumscribed Polygon** from the origin.
   Set the dimensions to **40 mm** (width) and **30 mm** (height) with **3 sides**. Finish the sketch.
3. **Extrude Polygon:**
4. Select **Extrude**, pick the polygon profile, set the thickness to **10 mm**, and press **Enter**.

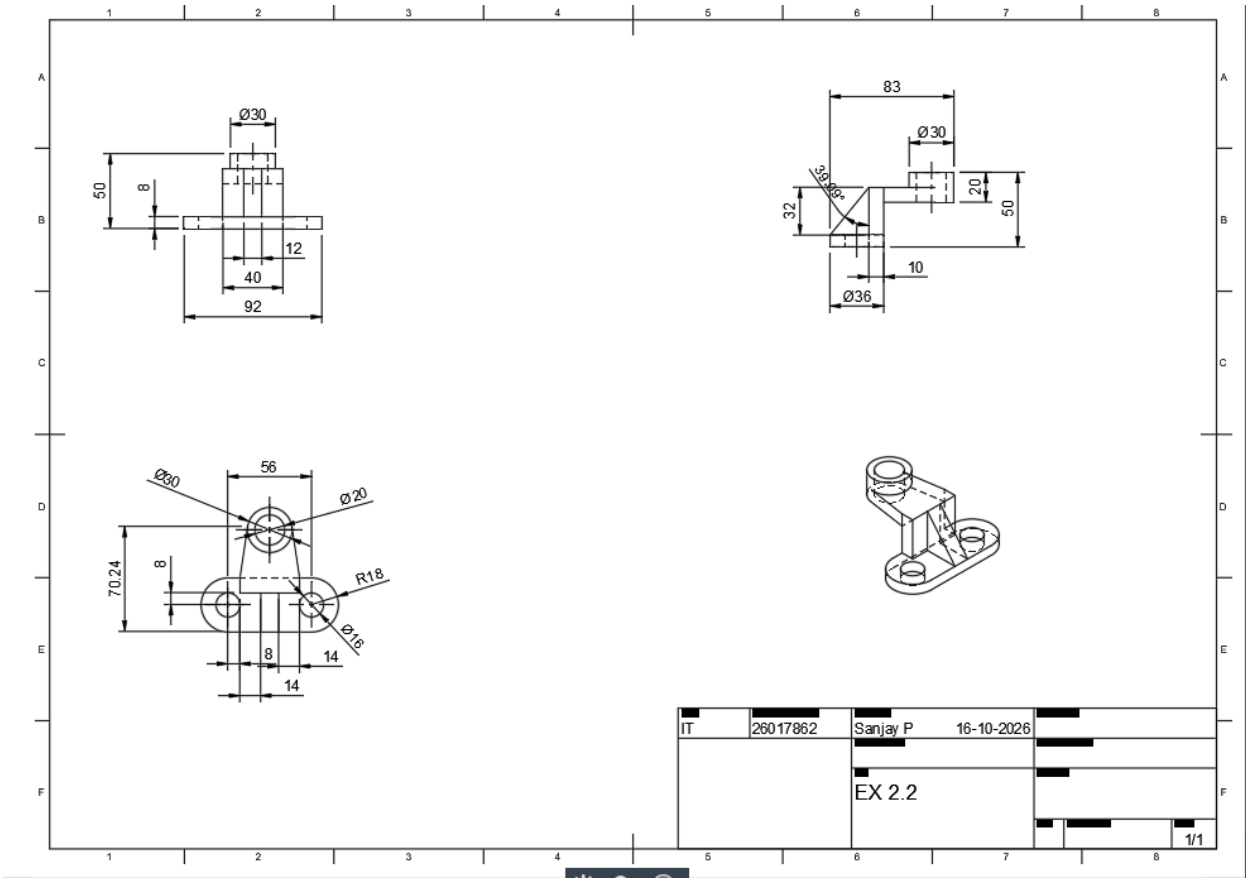
Step 5: Circular Collar Feature

1. **Concentric Circles:**
2. Create a new sketch on the polygon end face.
   Go to **Sketch > Circle > Center Circle** and place it at the reference center.
   Define an inner diameter of **dia 20 mm** and an outer diameter of **dia 30 mm**.
   Finish the sketch.
3. **Two-Side Extrude:**
4. Select **Extrude**, pick the circle profile, set the depth to **10 mm on both sides**, and press **Enter**.

INPUT:


(Ø30.00)
(Ø20.00)
(10.00)
(40.00)
(50.00)
(Ø16.00)
(10.00)
(R18.00)
-25
-50
-75
(R18.00)
(Ø16.00)
25
(22.00)
(12.00)
(22.00)

OUTPUT:



Result:

The 3D solid model has been successfully created from the given isometric drawing using Autodesk Fusion 360.